CIVITA · DIAGRAMA DE CLASES · VISTA HORIZONTAL

«external AndroidX»
ViewModel

01 · AUTENTICACIÓN

«class»
AuthViewModel

```
- repository: AuthRepository
- mutableState: MutableStateFlow<AuthUiState>
+ state: StateFlow<AuthUiState>
+ AuthViewModel(repository: AuthRepository)
+ email(email: String, password: String, register: Boolean): Unit
+ google(context: Context): Unit
+ reset(email: String): Unit
+ logout(context: Context): Unit
+ clearFeedback(): Unit
```

«class»
AuthRepository

```
- auth: FirebaseAuth
+ AuthRepository(auth: FirebaseAuth)
+ observeSession(): Flow<FirebaseUser?>
+ login(email: String, password: String): Unit {suspend}
+ register(email: String, password: String): Unit {suspend}
+ resetPassword(email: String): Unit {suspend}
+ google(context: Context): Unit {suspend}
+ logout(): Unit
+ clearGoogleSession(context: Context): Unit {suspend}
```

«data class»
AuthUiState

```
+ initializing: Boolean {readOnly}
+ uid: String? {readOnly}
+ displayName: String? {readOnly}
+ email: String? {readOnly}
+ busy: Boolean {readOnly}
+ error: String? {readOnly}
+ message: String? {readOnly}
```

«object»
AuthValidation

```
+ error(email: String, password: String, register: Boolean): String?
+ validEmail(email: String): Boolean
```

«class»
AppContainer

```
- app: FirebaseApp?
- auth: FirebaseAuth?
- db: FirebaseFirestore?
- storage: FirebaseStorage?
+ repository: ReportRepository?
+ authRepository: AuthRepository?
+ communityRepository: CommunityRepository?
+ AppContainer(context: Context)
```

02 · REPORTES

«class»
CivitaViewModel

```
- repository: ReportRepository
- mutableState: MutableStateFlow<CivitaUiState>
+ state: StateFlow<CivitaUiState>
- pendingDraft: ReportDraft?
- pendingId: String?
- observation: Job?
+ CivitaViewModel(repository: ReportRepository)
+ reload(): Unit
+ stopObserving(): Unit
+ selectCategory(id: String?): Unit
+ publish(draft: ReportDraft): Unit
+ clearPublication(): Unit
```

«interface»
ReportRepository

```
+ signIn(): String {suspend}
+ loadConfig(): AppConfig {suspend}
+ observeCategories(): Flow<RepositorySnapshot<IncidentCategory>>
+ observeReports(): Flow<RepositorySnapshot<UrbanReport>>
+ publishReport(draft: ReportDraft, requestId: String): String {suspend}
```

«class»
FirebaseReportRepository

```
- auth: FirebaseAuth
- firestore: FirebaseFirestore
+ FirebaseReportRepository(auth: FirebaseAuth, firestore: FirebaseFirestore)
+ signIn(): String {suspend}
+ loadConfig(): AppConfig {suspend}
+ observeCategories(): Flow<RepositorySnapshot<IncidentCategory>>
+ observeReports(): Flow<RepositorySnapshot<UrbanReport>>
+ publishReport(draft: ReportDraft, requestId: String): String {suspend}
```

«data class»
ReportDraft

```
+ categoryId: String {readOnly}
+ title: String {readOnly}
+ description: String {readOnly}
+ location: String {readOnly}
+ severity: Int {readOnly}
+ latitude: Double? {readOnly}
+ longitude: Double? {readOnly}
+ validationError(): String?
```

«object»
GeoHash

```
+ encode(latitude: Double, longitude: Double, precision: Int = 9): String
```

03 · COMUNIDAD

«class»
CommunityViewModel

```
- repository: CommunityRepository
- mutable: MutableStateFlow<CommunityUiState>
+ state: StateFlow<CommunityUiState>
- streams: Job?
- detail: Job?
- layer: Job?
- actionJob: Job?
- pendingComment: Triple<String,String,String>?
+ CommunityViewModel(repository: CommunityRepository)
+ start(): Unit
+ retry(): Unit
+ stop(): Unit
+ select(id: String): Unit
+ selectLayer(id: String): Unit
+ closeDetail(): Unit
+ clearFeedback(): Unit
```

«interface»
CommunityActions

```
+ saveProfile(name: String): Unit
+ preferences(radius: Long, ids: List<String>): Unit
+ comment(id: String, body: String): Unit
+ validate(id: String, value: String): Unit
+ support(id: String): Unit
+ flag(id: String, reason: String): Unit
+ moderate(id: String, status: String, reason: String): Unit
+ upload(id: String, context: Context, uri: Uri): Unit
+ markRead(id: String): Unit
+ subscribe(category: String, lat: Double, lon: Double, radius: Long): Unit
+ unsubscribe(id: String): Unit
+ evidenceUrl(path: String): String {suspend}
```

«class»
CommunityRepository

```
- auth: FirebaseAuth
- db: FirebaseFirestore
- storage: FirebaseStorage
+ CommunityRepository(auth: FirebaseAuth, db: FirebaseFirestore, storage: FirebaseStorage)
+ ensureProfile(): Unit {suspend}
+ isModerator(): Boolean {suspend}
+ observeProfile(): Flow<UserProfile?>
+ observePreferences(): Flow<UserPreferences>
+ saveProfile(name: String): Unit {suspend}
+ savePreferences(radius: Long, categoryIds: List<String>): Unit {suspend}
+ subscribe(categoryId: String, latitude: Double, longitude: Double, radius: Long): Unit
{suspend}
+ removeSubscription(id: String): Unit {suspend}
+ markRead(id: String): Unit {suspend}
+ events(): Flow<List<UrbanEvent>>
+ notifications(): Flow<List<UserNotification>>
+ subscriptions(): Flow<List<AlertSubscription>>
+ layers(): Flow<List<MapLayer>>
+ features(layerId: String): Flow<List<LayerFeature>>
+ comments(id: String): Flow<List<ReportComment>>
+ validations(id: String): Flow<List<ReportValidation>>
+ supports(id: String): Flow<List<ReportSupport>>
+ evidence(id: String): Flow<List<ReportEvidence>>
+ history(id: String): Flow<List<StatusChange>>
+ comment(id: String, body: String, requestId: String): Unit {suspend}
+ validate(id: String, value: String): Unit {suspend}
+ support(id: String): Unit {suspend}
+ flag(id: String, details: String): Unit {suspend}
+ moderate(id: String, status: String, reason: String): Unit {suspend}
+ uploadEvidence(id: String, context: Context, uri: Uri): Unit {suspend}
+ evidenceUrl(path: String): String {suspend}
```

04 · ESTADOS Y CUENTA

«data class»
CivitaUiState

```
+ loading: Boolean {readOnly}
+ categories: List<IncidentCategory> {readOnly}
+ reports: List<UrbanReport> {readOnly}
+ config: AppConfig? {readOnly}
+ uid: String? {readOnly}
+ selectedCategoryId: String? {readOnly}
+ fromCache: Boolean {readOnly}
+ error: String? {readOnly}
+ submitting: Boolean {readOnly}
+ publicationError: String? {readOnly}
+ publishedId: String? {readOnly}
+ visibleReports: List<UrbanReport> {readOnly, derived}
```

«data class»
CommunityUiState

```
+ profile: UserProfile? {readOnly}
+ preferences: UserPreferences {readOnly}
+ moderator: Boolean {readOnly}
+ events: List<UrbanEvent> {readOnly}
+ notifications: List<UserNotification> {readOnly}
+ subscriptions: List<AlertSubscription> {readOnly}
+ layers: List<MapLayer> {readOnly}
+ features: List<LayerFeature> {readOnly}
+ comments: List<ReportComment> {readOnly}
+ validations: List<ReportValidation> {readOnly}
+ supports: List<ReportSupport> {readOnly}
+ evidence: List<ReportEvidence> {readOnly}
+ history: List<StatusChange> {readOnly}
+ selectedId: String? {readOnly}
+ busy: Boolean {readOnly}
+ loadingDetail: Boolean {readOnly}
+ error: String? {readOnly}
+ message: String? {readOnly}
```

«data class»
AppConfig

```
+ allowedMediaTypes: List<String> {readOnly}
+ maxEvidenceBytes: Long {readOnly}
+ maxEvidenceCount: Int {readOnly}
+ minRadiusMeters: Long {readOnly}
+ maxRadiusMeters: Long {readOnly}
```

«data class»
UserPreferences

```
+ alertRadiusMeters: Long {readOnly}
+ categoryIds: List<String> {readOnly}
+ pushEnabled: Boolean {readOnly}
+ preciseLocationAllowed: Boolean {readOnly}
```

«data class»
ReportComment

```
+ id: String {readOnly}
+ authorId: String {readOnly}
+ authorName: String {readOnly}
+ body: String {readOnly}
+ hidden: Boolean {readOnly}
```

05 · DOMINIO / ANDROID

«class»
MainActivity

```
# onCreate(savedInstanceState: Bundle?): Unit
```

«class»
ComponentActivity

«data class»
IncidentCategory

```
+ id: String {readOnly}
+ label: String {readOnly}
+ color: Color {readOnly}
+ defaultDurationHours: Long {readOnly}
```

«data class»
UserProfile

```
+ id: String {readOnly}
+ fullName: String {readOnly}
+ email: String {readOnly}
+ role: String {readOnly}
+ status: String {readOnly}
+ trustLevel: Long {readOnly}
```

«data class»
UrbanReport

```
+ id: String {readOnly}
+ title: String {readOnly}
+ categoryId: String {readOnly}
+ category: String {readOnly}
+ description: String {readOnly}
+ location: String {readOnly}
+ severity: Int {readOnly}
+ time: String {readOnly}
+ status: ReportStatus {readOnly}
+ confidence: Int {readOnly}
+ color: Color {readOnly}
+ creatorId: String {readOnly}
```

«enumeration»
ReportStatus

```
Review
Verified
Resolved
+ label: String
+ documentValue: String
+ fromDocument(value: String): ReportStatus {companion}
```